

UC Name	<i>UC-01: Manage User Accounts</i>
Summary	<i>Administrator creates, updates, deactivates user accounts and assigns roles/permissions.</i>
Dependency	<i>None</i>
Actors	<i>Primary: Administrator</i>
Preconditions	<i>Administrator is authenticated; system is accessible; administrator has user-management permission.</i>
Description of the Main Sequence	• <i>Step 1: Administrator logs in.</i> • <i>Step 2: System authenticates credentials.</i> • <i>Step 3: Administrator opens User Management.</i> • <i>Step 4: System loads current user records.</i> • <i>Step 5: Administrator enters user data.</i> • <i>Step 6: System validates data.</i> • <i>Step 7: System checks for duplicates.</i> • <i>Step 8: Administrator assigns roles and permissions.</i> • <i>Step 9: System saves user account.</i> • <i>Step 10: System saves assigned roles/permissions.</i> • <i>Step 11: Confirmation message is displayed.</i>
Description of the Alternative Sequence	• <i>Step 1: Invalid user data entered → system shows error</i> • <i>Step 2: Duplicate user detected → system prevents creation</i> • <i>Step 4: Role/permission missing → system requests correction.</i> • <i>Step 4: Save failure → system displays failure message.</i>
Non functional requirements	<i>Secure authentication</i> <i>Role-based access control</i> <i>Fast response time</i>
Postconditions	<i>User account is successfully/created/updated/deactivated</i>

UC Name	<i>UC-02: Configure System Settings</i>
Summary	<i>Administrator views and modifies system configuration settings.</i>
Dependency	<i>UC-01</i>
Actors	<i>Primary:</i> <i>Administrator</i>
Preconditions	<i>Administrator is logged in; administrator has configuration access.</i>
Description of the Main Sequence	• <i>Step 1: Administrator logs in.</i> • <i>Step 2: Open Configuration Settings.</i> • <i>Step 3: System loads current settings from repository.</i> • <i>Step 4: Administrator modifies parameters.</i> • <i>Step 5: System validates settings.</i> • <i>Step 6: Valid settings are applied to the system.</i> • <i>Step 7: Updated settings are saved.</i> • <i>Step 8: Confirmation message is shown.</i>
Description of the Alternative Sequence	• <i>Step 1: Invalid configuration → system rejects input</i> • <i>Step 2: Repository save failure → system keeps previous settings.</i>
Non functional requirements	<i>System stability</i> <i>Configuration validation</i>
Postconditions	<i>System settings are updated and stored successfully.</i>

UC Name	<i>UC-03: Monitor System Logs</i>
Summary	<i>Administrator searches, filters, views, and reviews system activity logs.</i>
Dependency	<i>None</i>
Actors	<i>Primary: Administrator</i>
Preconditions	<i>Administrator is authenticated; logs exist or log repository is accessible.</i>
Description of the Main Sequence	• <i>Step 1: Administrator logs in.</i> • <i>Step 2: Open Logs Dashboard.</i> • <i>Step 3: System loads the dashboard.</i> • <i>Step 4: Administrator enters filter/search criteria.</i> • <i>Step 5: System retrieves matching logs.</i> • <i>Step 6: System displays log list.</i> • <i>Step 7: Administrator selects a log.</i> • <i>Step 8: System loads detailed log information.</i> • <i>Step 9: Administrator reviews activity logs.</i>
Description of the Alternative Sequence	• <i>Step 1: No logs found → system shows empty result</i> • <i>Step 2: Invalid filter → system asks for new criteria.</i> • <i>Step 3: Repository unavailable → system shows retrieval error.</i>
Non functional requirements	<i>High performance log retrieval Secure log storage</i>
Postconditions	<i>Logs are reviewed or no matching logs are reported.</i>

UC Name	<i>UC-04: Manage Airport Reference Data</i>
Summary	<i>Administrator manages airport reference data such as gates, stands, airport codes, and operational reference records.</i>
Dependency	<i>UC-02</i>
Actors	<i>Primary: Administrator</i>
Preconditions	Administrator is logged in; reference data module is available.
Description of the Main Sequence	• <i>Step 1: Administrator logs in.</i> • <i>Step 2: Accesses Reference Data Module.</i> • <i>Step 3: System loads current reference data.</i> • <i>Step 4: Administrator adds, updates, or deletes data.</i> • <i>Step 5: System validates data.</i> • <i>Step 6: Valid changes are applied.</i> • <i>Step 7: Updated reference data is saved.</i> • <i>Step 8: Confirmation message is shown.</i>
Description of the Alternative Sequence	• <i>Step 1: Invalid data → system shows error</i> • <i>Step 2: Duplicate reference value → system rejects change.</i> • <i>Step 3: Delete blocked due to dependency → system displays warning.</i>
Non functional requirements	<i>Data integrity</i> <i>Validation rules</i>
Postconditions	<i>Reference data is updated successfully.</i>

UC Name	<i>UC-05: Assign Gates to Flights</i>
Summary	<i>Operations Staff assigns available gates to scheduled flights while avoiding conflicts.</i>
Dependency	<i>UC-06</i>
Actors	<i>Primary: Operations Staff</i>
Preconditions	<i>Flight schedule exists; gates exist; system can access flight and gate repositories.</i>
Description of the Main Sequence	• <i>Step 1: Operations Staff views flight schedule.</i> • <i>Step 2: System loads scheduled flights.</i> • <i>Step 3: Operations Staff selects a flight.</i> • <i>Step 4: System retrieves flight details.</i> • <i>Step 5: System retrieves available gates.</i> • <i>Step 6: System validates gate assignment.</i> • <i>Step 7: Operations Staff confirms selected gate.</i> • <i>Step 8: System creates assignment.</i> • <i>Step 9: Flight gate status is updated.</i> • <i>Step 10: Assignment is saved.</i> • <i>Step 11: Confirmation message is shown.</i>
Description of the Alternative Sequence	• <i>Step 1: Gate unavailable → system prompts another selection</i> • <i>Step 2: Assignment conflict → system displays alternative gates.</i> • <i>Step 3: No gates available → system informs staff.</i> • <i>Step 4: Save failure → assignment is not confirmed.</i>
Non functional requirements	<i>Real-time updates</i> <i>Conflict prevention</i>
Postconditions	<i>Gate is assigned to the flight and stored successfully.</i>